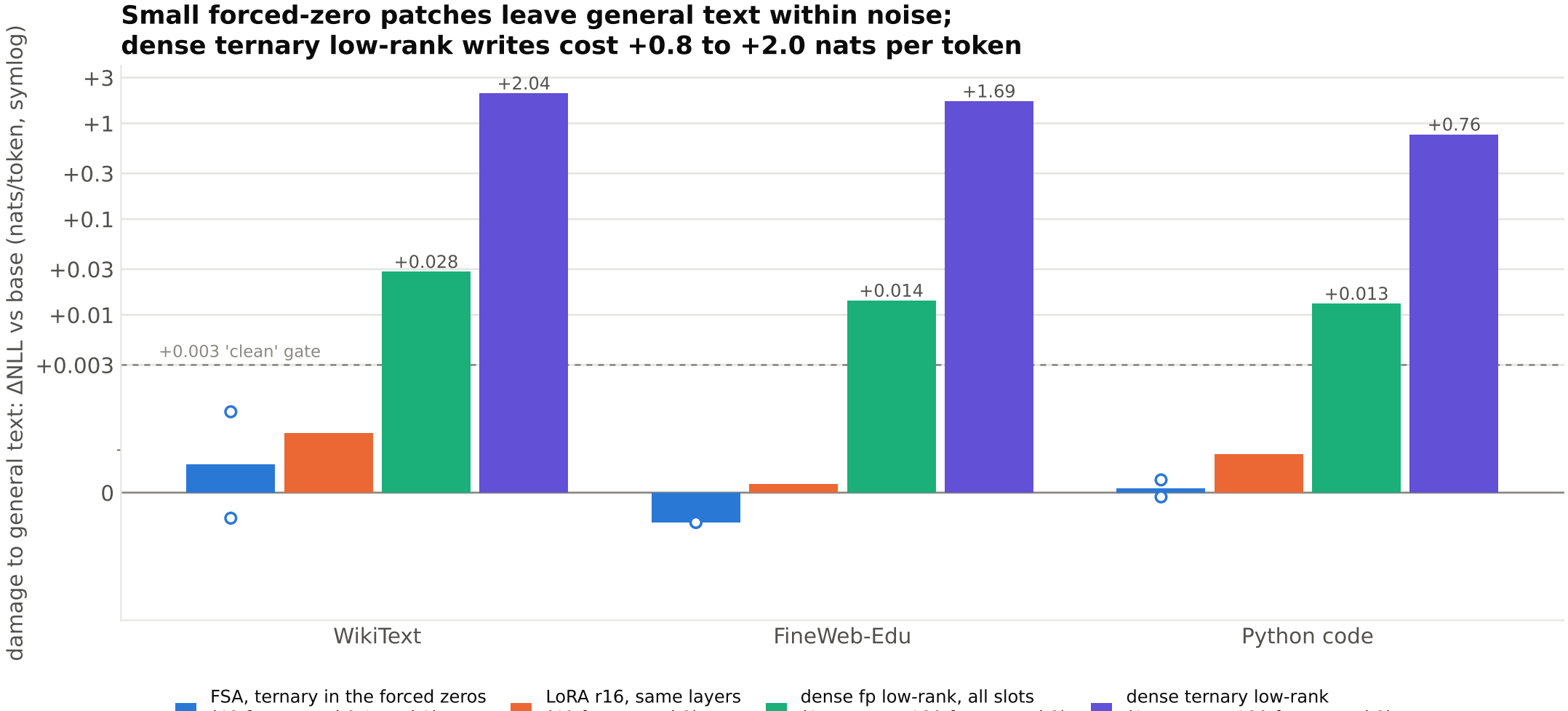


ΔNLL on held-out windows (40×512 WikiText; FineWeb-Edu and Python code panels), paired SE ≈ 0.002, so FSA and LoRA on 40 facts (MoE layers 14,15, Bop / LoRA recipes) are within noise of zero. The dense cells are the interim 2×2 factorial (1 person × 100 facts, matched Adam recipe, val-selected LR, one seed; sparse cells still running): dense ternary reaches 100% test recall at +2.0 nats; dense fp was LR-limited at 200 steps (17.5% recall). Larger forced-zero patches are not free: the 3p×100 patches are +0.011 (4M) and +0.007 (16M) on code, above the gate.